

ORIGINAL RESEARCH

Experimentation framework for wireless communication systems under jamming scenarios

Marko Jacovic¹  | Michael J. Liston¹ | Vasil Pano¹ | Geoffrey Mainland² | Kapil R. Dandekar¹

¹Department of Electrical and Computer Engineering, Drexel University, Philadelphia, Pennsylvania, USA

²Department of Computer Science, Drexel University, Philadelphia, Pennsylvania, USA

Correspondence

Kapil R. Dandekar, Department of Electrical and Computer Engineering, Drexel University, 3141 Chestnut Street, Philadelphia, PA 19104, USA.
Email: krd26@drexel.edu

Funding information

National Science Foundation, Grant/Award Numbers: CCF-1717088, CNS-1730140; U.S. Department of Education, Grant/Award Number: P200A180082

Abstract

Cyber-physical systems (CPS) integrate control, sensing, and processing into interconnected physical components to support applications within transportation, energy, healthcare, environment, and various other areas. Secure and reliable wireless communication between devices is necessary to enable the widespread adoption of these emerging technologies. Cyber-physical systems devices must be protected against active threats, such as Radio Frequency (RF) Jammers, which intentionally disrupt communication links. Jamming detection and mitigation techniques must be evaluated extensively to validate algorithms prior to full implementation. Challenges related to obtaining zoning permits, Federal Aviation Administration (FAA) pilot certification for Unmanned Aerial Vehicles (UAVs), and Federal Communications Commission (FCC) licencing lead to evaluation limited to simulation-based or simplistic, non-representative hardware experimentation. A site-specific ray-tracing emulation framework is presented to provide a realistic evaluation of communication devices under RF jamming attacks in complex scenarios involving mobility, vehicular, and UAV systems. System architecture and capabilities are provided for the devices under test, real-world jamming adversaries, channel modelling, and channel emulation. Case studies are provided to demonstrate the use of the framework for different applications and jamming threats. The experimental results illustrate the benefit of the ray-tracing emulation system for conducting complex wireless communication studies under the presence of RF jamming.

KEYWORDS

cyber-physical systems, hardware-in-the-loop stimulation, telecommunication security

1 | INTRODUCTION

With the emergence of new wireless technologies in our everyday lives, such as in Cyber-Physical Systems (CPS), Vehicular, Unmanned Aerial Vehicle (UAV) communications, and the Internet of Things (IoT), protecting devices against external threats is essential for society to thrive in the future. Cyber-Physical Systems in particular has a high dependence on communication systems for data and control, leading to vulnerabilities to attacks [1]. Radio frequency (RF) jammers disrupt the wireless RF medium by transmitting signals with the malicious intent to interfere with legitimate communication

systems. In critical communication systems, for instance, autonomous vehicles, medical device readings, and mobile access points for disaster relief, the consequences can be catastrophic.

Types of RF jammers include constant, random, intermittent, reactive, and intelligent [2]. The constant jammer is straightforward in operation, sending a jamming signal at a centre frequency with a given bandwidth. The continuous nature of the threat results in simple detection and inefficient energy usage; therefore, more complex jammers are implemented. Random jammers trigger a jamming event with a set probability; similarly, intermittent jammers operate in set

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time intervals. Both random and intermittent variants are incremental improvements over the constant jammer while more advanced techniques are utilised by reactive and intelligent jammers. Reactive jammers sense activity in the RF medium prior to transmitting a jamming signal and intelligent jammers leverage knowledge of protocols used to surgically target specific components of the system [2]. In this work, all the aforementioned jamming types are considered as threats to legitimate communication systems.

Defence against jamming attacks includes detection of a threat and implementation of mitigation techniques to attempt to circumvent the interference source [2, 3]. More advanced techniques use the classification of jammer types to help select the appropriate method of mitigation [3].

An inherent challenge exists when evaluating the impact of jamming as well as classification and mitigation techniques under complex emerging wireless communication scenarios, such as the CPS, IoT, and other examples mentioned previously. In these cases, the scenarios are difficult to examine with hardware in the loop due to mobility, zoning restrictions, scalability of nodes, repeatability of testing, and of course the legality of jamming the RF medium. Previous assessments of jamming defence techniques have focussed predominately on simulation [4–10] and analytical solutions [1, 11–13], both of which ignore practical constraints associated with hardware and RF impairments.

A methodology must be developed to realistically evaluate the aforementioned communication scenarios against different active threats without threatening real systems. We propose a platform suitable for the evaluation of a diverse array of communication systems under attack by various types and numbers of RF jammers. The focal point of our framework consists of a bridge connecting ray-tracing software to a highly scalable channel emulator, which can be connected to either flexible software-defined radio (SDR) or commercial off-the-shelf (COTS) CPS communication links. Ray tracing has been shown through experimental validation to be an accurate tool for site-specific models [14, 15]. In addition, ray tracing has been used in combination with measurement campaigns for 5G third Generation Partnership Project (3GPP) mmWave channel model development [16] and for time alignment of statistical mmWave models [17]. We use ray tracing to develop realistic site-specific channels, which vary over time and are used to program the emulator, such that external hardware consisting of legitimate devices and jamming sources may be considered for analysis under complex scenarios. The main contributions of this work are as follows: (i) our testbed enables the safe use and evaluation of real-world RF jammers with communication systems, which otherwise would be considered infeasible, (ii) the proposed framework provides the flexible generation of new test scenarios with various attributes of mobility, environment, and antenna properties, (iii) furthermore, the overall workflow and methodology we present are straightforward to extend to other channel emulators.

The paper is organised as follows. Related work is discussed in Section 2. System components and capabilities are

described in Section 3. Example case studies using the framework are provided in Section 4. Discussions related to the testbed and ongoing efforts are presented in Section 5 and concluding remarks are provided in Section 6.

2 | RELATED WORK

We presented our Grid SDR Network Testbed for measurement and emulation in [19]. The testbed consists of an elevated grid housing for a combination of over 20 Ettus Universal Software Radio Peripheral (USRP) X310 and N210 radios, each with over-the-air and wired RF capabilities into the EchoRidge Dynamic Spectrum Environment Emulator (DYSE) [29], and Dragon Radio [19], an open-source¹ Orthogonal Frequency Division Multiplexing (OFDM)-based SDR with highly flexible Medium Access Control (MAC), link, and network layers. Our infrastructure includes 12 Ettus USRP X310 SDRs with direct access to the emulator. Using a Linux Container (LXC)-based system for managing radio experiments compatible with the NSF Platforms for Advanced Wireless Research (PAWR) Colosseum [20], we have similarly made our Grid SDR Network Testbed available for external experimentation.

Constant, random, and intermittent jammers are relatively simple and may be implemented using open-source additive white Gaussian noise (AWGN) signal generators [21].

Decision fusion with multiple antennas was used to mitigate the effects of constant jamming against sensor networks [18]. Sub-optimal approaches were presented to reduce complexity and validated through numerical and simulation results, which may be further enriched with different threat models and experimental analysis.

Reactive jammers are inherently more difficult to implement; they must sense the medium, determine the presence of a transmitting device, and then jam the legitimate system prior to the conclusion of an active communication event. In this work, we make use of a reactive jammer developed by us and reported in [22]. This reactive jammer is described further in Section 3.2.2 and has the following key specifications: run-time jamming parameter configuration, rapid response time, and ability to use different transmit signals during operation. A reactive jammer for future Cross-Technology Communication (CTC) systems was developed in [23]. Proposed CTC systems will convey information between heterogeneous devices by changing energy levels or packet duration size. The jammer implemented on an Ettus USRP N210 SDR is capable of responding on an average of 28 ms, which is sufficient for the specific designed CTC scenario; however, this may not be sufficient for most scenarios or for homogeneous systems. In comparison, our reactive jammer [22] is capable of reacting within 80 ns. A network of reactive jammers was built upon the Linksys WRT54GL Access Point to perform combined friendly jamming against a detected adversary in [24].

¹<https://drexelwireless.github.io/dragonradio/>

Experiments in an indoor office setting demonstrated favourable results; however, the studies could be enriched by a flexible radio format in comparison to the static WRT54GL, and the testing of complex communication scenarios. Our reactive jammer is capable of adjusting various jamming attributes such as duration, delay, and signal type. In addition, our proposed system, described further in Section 3, provides a methodology for testing various communication scenarios of different complexity. A protocol-aware intelligent jammer was able to conserve energy in comparison to a continuous jammer by targeting the network and transport layers [25]. Jamming events were triggered based on offline trained features of detected packet size, timing, and sequence of message types. The evaluation was performed using the ZigBee routing protocol and did not consider the realistic requirements of the MAC layer. Software-defined radios were used to compare the effect of tone, sweeping, and protocol-aware jamming signals for UAV control systems in [26]. Detection was based on full demodulation of received signals while the protocol-aware jamming signals consisted of a separate instance of the legitimate transmitter, both requirements are unrealistic conditions. Our reactive jammer from the work in [22] is capable of targeting IEEE 802.11 systems using the Distributed Coordination Function MAC without any additional requirements in contrast to the works in [25, 26]. We provide a case study highlighting this functionality in Section 4.3. All of the above systems were only experimentally tested for a single link in an attenuated wired setup, which cannot provide the realistic channel and interference scenarios discussed in this paper.

An intelligent jammer for IEEE 802.11 signals was developed to target specific parts of the preamble to corrupt carrier frequency offset estimation in [27]. Legitimate performance was shown to degrade substantially; however, an experimental evaluation was limited to two simple 3-node scenarios and the response time of the jammer was on the order of milliseconds, requiring substantial timing assumptions for the test setup. Our reactive jammer as mentioned previously has a rapid response time of within 80 ns from detection. Again, the experimental studies of the work in [27] demonstrate the challenges in evaluating jamming in realistic scenarios, thus motivating the testbed described in this paper. Constant, random, deceptive (a periodic variant), and reactive jammers were implemented against sensor networks in [28]. Detection methods were paired with frequency and spatially agile techniques to mitigate the effects of the aforementioned threats. The experimental results discussed in greater detail in [3] may be further enriched by considering higher complexity devices (e.g. Wi-Fi) than the MICA2 Mote platform and scenarios (e.g. outdoor, mobility) for the evaluation.

We have provided a summary of the related works in Table 1. Existing works have been limited due to the following technical challenges that we overcame in the development of our framework: (i) ray-tracing software often is limited to static physical environments and requires additional development for moving objects, (ii) the conversion of ray-tracing simulation data to a format usable by a channel emulator must be automated for practical use with a large number of channel profiles,

and (iii) channel emulators require calibration and configuration routines that are often tedious to implement. In addition, with the exception of studies using the PAWR Colosseum, it has largely been infeasible for existing works to consider a scalable number of full-stack SDRs and jamming sources with a channel emulator capable of supporting the physical connections required. We leverage our Grid SDR Network Testbed for the infrastructure required to enable our studies.

3 | SYSTEM ARCHITECTURE AND CAPABILITIES

3.1 | Experiment methodology

Our software-defined radio (SDR) testbed [19] enables repeatable and scalable testing of SDRs in both wireless and wired environments using a 24×24 RF environment channel emulator. The wired configuration allows for SDRs to be connected through the full-mesh 24×24 EchoRidge DYSE channel emulator, enabling highly scalable, repeatable and controllable testing of complex scenarios with hardware-in-the-loop [29]. Wireless testing can also be performed using over the air testing in an indoor laboratory environment using COTS or custom antennas. The testbed has over 30 grid nodes, which consist of an Ettus USRP X310 or N210 SDR with a dedicated host computer connected via Ethernet. We have used Wireless Insite, ray-tracing software developed by Remcom, Inc. [30], to model wireless channel conditions that are inherently difficult to study such as UAV Air-to-Ground data links [19, 31]. The use of Wireless Insite with our testbed enables a realistic highly scalable evaluation of devices under scenarios with high mobility, unique height profiles, urban and outdoor settings, and carrier frequencies up to 100 GHz. Various devices may be tested in a controlled environment under the presence of an active jamming attack through the use of our infrastructure. In addition to our SDRs, commercial devices may also be integrated into the testbed as devices under test. Linux containers with constant, periodic, random, and reactive jammers are all available as SDR-based threats for the devices under test in the testbed. DragonJammer, a periodic and random RF jammer, is used to support the threat options as well as the reactive jammer described previously in [22]. An overview of our proposed jamming testbed is shown in Figure 1.

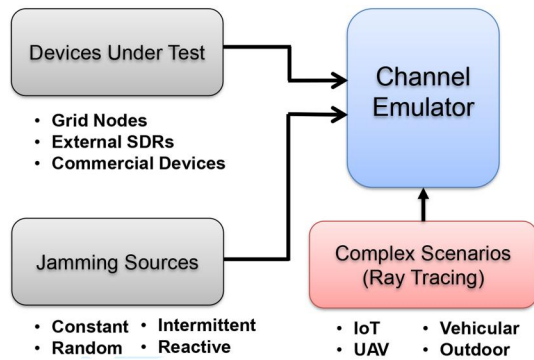
3.2 | Devices under test

3.2.1 | Legitimate communication systems

The proposed jamming testbed supports a wide variety of communications systems including DragonRadio, an open-source, full radio stack implemented purely in software, which runs on Ettus USRP SDRs. DragonRadio features Frequency-division multiple access (FDMA), Time-division multiple access (TDMA), ALOHA and hybrid MAC protocols as well as all Physical (PHY) layers offered in LiquidDSP [32].

TABLE 1 Summary of works related to the proposed testbed

Work	Focus	Comparison
[1, 4–13, 18]	Jamming mitigation techniques	Limited to numerical and simulation results
[19, 20]	Scalable SDR emulation testbed	Enabling hardware infrastructure, limited complexity of emulation scenarios
[21]	Open-source signal generator	Enabling technology for simple SDR jammers
[22]	Our protocol-aware reactive jammer	Used as a jamming source in our testbed
[23]	CTC reactive jammer on USRP N210	Long response time in comparison to our reactive jammer, may not be applicable for homogenous systems, single link considered
[24]	Network of reactive jammers using Linksys WRT54GL	Jammer has limited configurable jamming attributes, experiments only in indoor setting, single link considered
[25]	Intelligent jammer targeting network and transport layers	Jamming events triggered offline, MAC layer was ignored, single link considered
[26]	Protocol-aware jamming of UAV control signals	Requires full demodulation of received signals in contrast to our reactive jammer which requires only the preamble, single link considered
[27]	Intelligent jammer targeting CFO estimation portion of IEEE 802.11 preamble	Response time on order of ms in comparison to our 80 ns, experimental evaluation limited to three nodes
[3, 28]	Sensor network implementation of various jammer threats, detection, and mitigation techniques	Only low-complexity devices considered, limited indoor experimental evaluation

**FIGURE 1** Jamming testbed overview. High-level system components to enable experimentation

Changing between different configurations of MAC protocols and PHY layers may be done programmatically or via an embedded Python interpreter. DragonRadio scales to multi-node networks making it a useful test system for the jamming testbed described in this paper. Commercial communications systems are also supported given the devices have accessible transmit/receive (TX/RX) coax antenna ports for compatibility with our channel emulator. Devices with a single-antenna port may be connected using an RF circulator.

3.2.2 | Jamming adversaries

A wide array of attack types must be considered to develop and evaluate robust methods for jamming classification and mitigation. A constant jammer is easily implemented using a

signal generator from the open-source GNU radio library. We have developed DragonJammer to create both periodic and random jamming signals. A high-level overview of its operation is provided in Figure 2. DragonJammer is developed in C++ and built upon the LiquidDSP library and uses chirp signals as the transmitted waveform [32]. External settings configured at the command line are shown in Figure 2 in orange. A USRP Hardware Driver (UHD) object is initialised with a desired sample rate, transmit gain, and carrier frequency.

The chirp signal is generated based on the sample rate, bandwidth of the signal, total time of the chirp, and the dwell time of the signal during the sweep. The jammer transmits the chirp signal from the UHD object based on the variable burst duration and off duration. The periodic jammer has an off period equal to the rest duration. In contrast, after each jamming event, the random jammer selects the off period based on a random integer within a configured range.

We previously built a real-time, protocol-aware reactive jammer that executes on the Field-programmable gate array (FPGA) fabric of an SDR [22]. The system is capable of configuring various jamming- and radio-related parameters in run-time operation. Vital jamming-related specifications such as gain, duration, additional delay, signal type, detection thresholds, and reference sequences may all be set while the system is active and operating. An overview of the reactive jammer receiver and transmitter chains is provided in Figure 3 with configurable parameters shown in orange. Both calculations are compared to separate thresholds, which are used to detect that a signal is present on the medium. The transmitter is capable of sending the buffered received samples as a replay attack, AWGN, or a stored transmit buffer.

FIGURE 2 DragonJammer overview. Random and periodic jammer using chirp signals

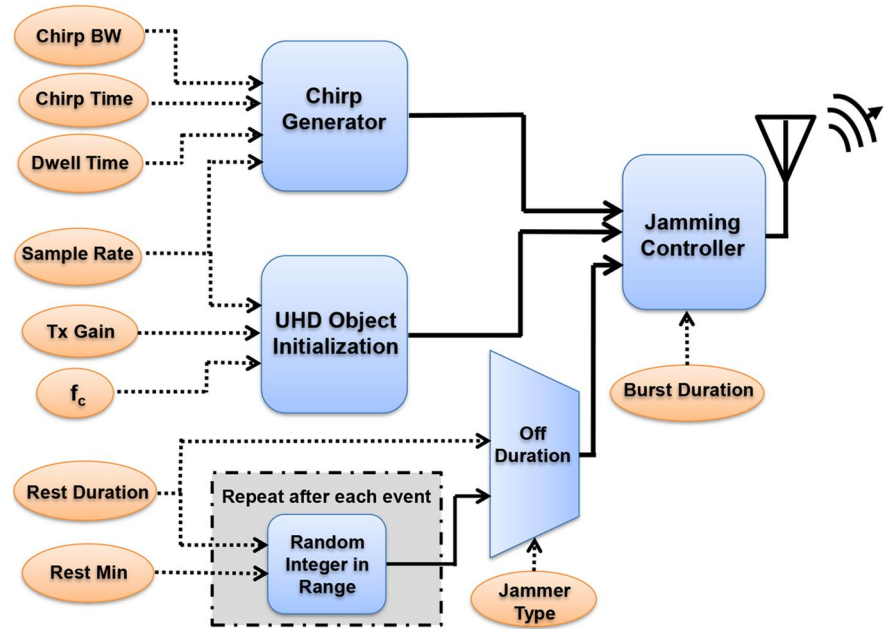
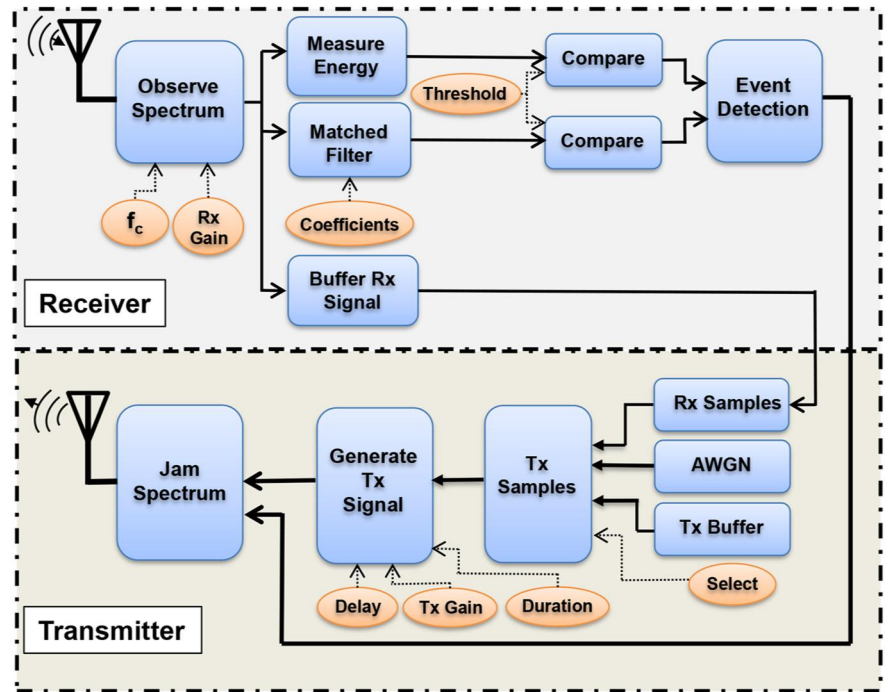


FIGURE 3 Reactive jammer overview



The duration, gain, and delay of the transmit signal are also configured prior to jamming the spectrum. A key enabling aspect of the jammer is that signal detection and transmission are abstracted to the hardware components of the SDR, resulting in a rapid RF response time within 80 ns of detection. All of the aforementioned jamming adversaries are implemented on SDR and capable of disrupting real communication systems. The proposed system provides a controllable and safe approach to testing with the RF jammers.

3.3 | Channel modelling and emulation

Existing methods for the evaluation of RF jamming scenarios are limited primarily to (i) hardware in simple indoor settings, offering only a sub-set of future and present RF technologies; and (ii) simulation of complex situations, which lack real RF impairments and bypass the requirements associated with physical radios. At the core of our jamming testbed, we have developed a framework for bringing ray-traced simulated channels to a large-scale channel emulator, which can be

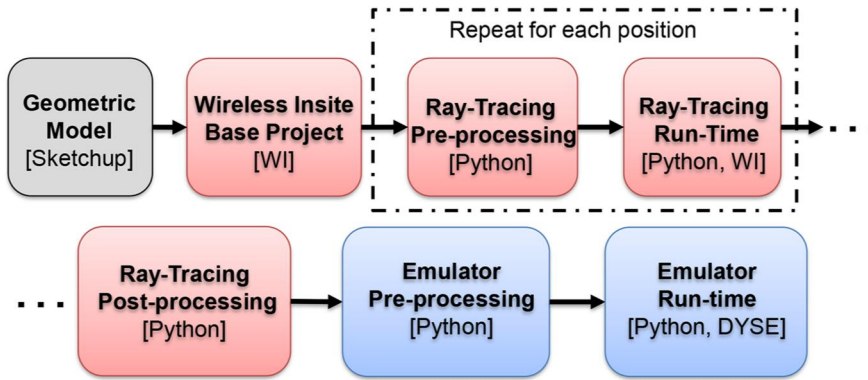


FIGURE 4 Complex scenario development. Software design flow combining ray-tracing studies with a channel emulator

applied to real hardware, enabling us to avoid the aforementioned common pitfalls of a traditional wireless communication systems evaluation. Ray-tracing tools are used to characterise propagation environments that are inherently challenging to model. A summary of the key components of the software design flow of the complex scenario development is shown in Figure 4. Our design process begins by selecting a geometric model to represent an area of interest; we choose to use Sketchup for its large open-source library of models in the 3D Warehouse as well as for its compatibility with our ray-tracing software [33]. The geometric model is imported into Wireless Insite as various features with unique defined materials. A summary of the different electrical properties of the common materials used is provided in Table 2. Transceiver sets, waveforms, study area, and desired outputs are defined in the project. The base files generated by the Wireless Insite project are modified for any object or transceiver movement through our Python codebase. The simulations are executed at each time step with corresponding output data from the studies. Post-processing is performed to format the Wireless Insite output into files compatible with the DYSE unit, however, which may be modified for use with other channel emulators. Additional scripts are used to calibrate the physical devices under test, extend the study if required, and re-map nodes to different DYSE channels. Control of the DYSE emulator and the generation of configuration files is automated through the codebase.

3.3.1 | Channel modelling

Wireless Insite supports the import of various Computer-Aided Design (CAD) file types and Keyhole Markup language Zipped (KMZ) files based on Google Earth. We first select a base model of a specific building or area from the 3D Warehouse and create a new project. Then, we use CAD-MAPPER, a CAD file generator that uses public source map data, to import surrounding buildings, roads, parks, and water bodies into the project [34]. Geometric objects with high complexity are simplified to reduce computational complexity. Single studies are run with and without the modifications to verify that the simplification does not alter results. All objects in the Sketchup model are separated by material, then exported

TABLE 2 Electrical properties of materials used in Wireless Insite

Material	Conductivity (Si/m)	Relative permittivity
Asphalt	5×10^{-4}	5.72
Brick	1×10^{-3}	4.44
Concrete	15×10^{-3}	7.00
Dry Earth	1×10^{-3}	4.00
Metal	PEC	∞
Water	22×10^{-2}	81.00

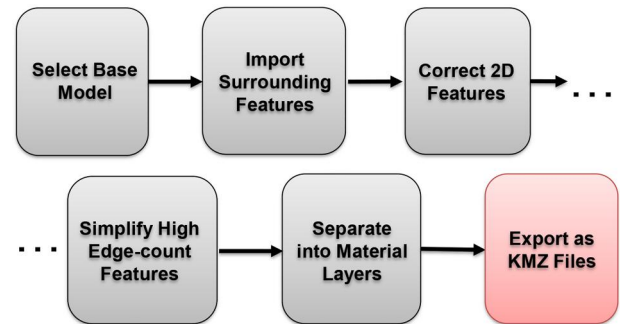


FIGURE 5 Geometric base model development through Sketchup

individually in a KMZ format for Wireless Insite. A summary of the geometric model process using Sketchup is shown in Figure 5. In Wireless Insite, each KMZ file is imported as a feature to the project. Corresponding materials for each feature are defined using the existing feature library and user-defined properties if necessary. We use sinusoid waveforms and omni-directional antennas with a carrier frequency and signal bandwidth of interest for the study. Transceiver sets are defined for each legitimate communication system and jammer in the experiment. In addition, we use transceiver sets to define moving vehicle trajectories and node movement area boundaries to be used later in our Python pre-processing code-base. We define a study area to encompass the close surrounding area of all active transceivers in the study. We use the accurate X3D Ray Propagation Model provided by Wireless Insite for optimised path calculations using graphics processing unit (GPU) acceleration and multi-threading of core processors.

The requested output consists of the channel impulse response (CIR), direction of arrival (DOA), and direction of departure (DOD) between each transceiver in the study. The direction of arrival and DOD are optional outputs that are only required if directional antenna gains are applied to the CIR in post-processing. An overview of the design flow for the Wireless Insite base project is shown in Figure 6.

Pre-processing of the Wireless Insite files enables a dynamic transceiver and object movement, which are not feasible through working only with the GUI. In all Python-based stages of our design flow from Figure 4, we use JavaScript Object Notation (JSON) configuration files to parameterise different study options. The pre-processing JSON options are primarily based on movement, transceiver identification, study duration, and output. The components of the Wireless Insite pre-processor and execution code are shown in Figure 7. The movement of objects and transceivers in the study is optional; however, positional change of at least one component results in a more interesting time-varying scenario. The transceiver definition file, with extension type **.txrx*, is used to extract: (i) the initial coordinates of the transceivers, (ii) the network boundaries for random movement of the physical nodes, and (iii) any active vehicle trajectories. Object definition files, with extension type **.object*, are selected from the JSON file to determine: (i) a valid surface area used for height extraction, (ii) a moving base object such as a vehicle, and (iii) a static base

object to be copied over a set of points. The static object is duplicated and translated over a transceiver trajectory set with configurable separation between copies. The moving object transceiver trajectory is used to create a target path for the object to travel over. At each time step, the valid surface area is used to update the new moving object height and the corresponding Wireless Insite files. Nodes may be configured for random movement within a bounded area defined by positional markers set using an inactive transceiver set. The transceiver points move in random directions for a configurable length of time prior to changing to a new direction. If nodes move within a configurable minimum separation distance or past the network boundaries, a new random direction is applied. Random seeds may be set from the JSON configuration file to ensure repeatability of the studies. The valid surface area object is also used to update the height for each transceiver point. The Wireless Insite project is run from the command line after the files are updated, and the generated output files are stored into sub-directories representing each time step.

3.3.2 | Emulation

The output files from Wireless Insite are post-processed into a format supported by the DYSE channel emulator, in which

FIGURE 6 Wireless Insite base project design flow

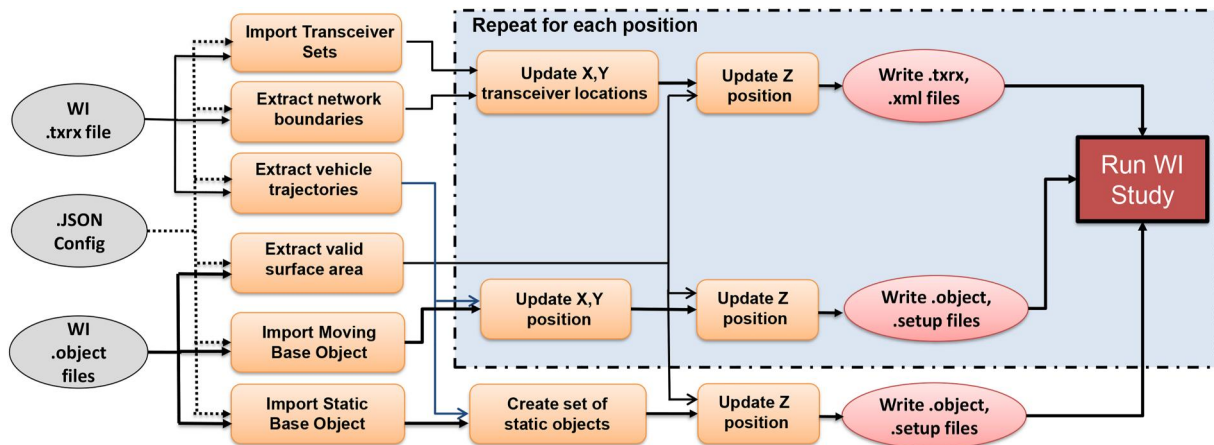
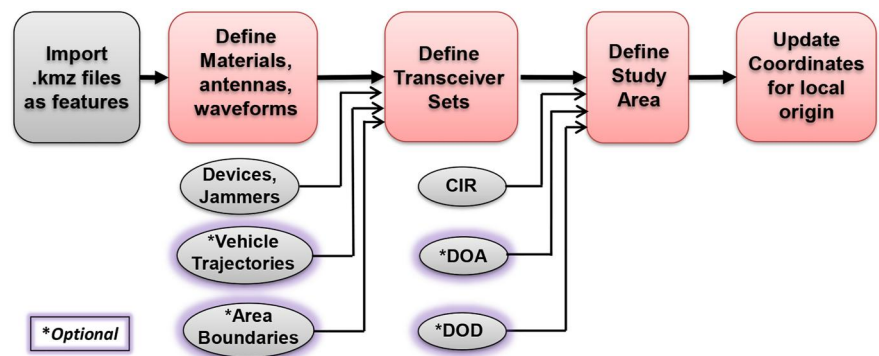


FIGURE 7 Wireless Insite pre-processing and execution

channels between each transceiver pair are defined at every channel update timestamp. An important consideration is that the channel modelling described in the previous section may be applied to various channel emulators and is not restricted to only DYSE. A JSON configuration file is used to define active devices under test and jammers by name from the Wireless Insite project. The Wireless Insite transceiver names and ID numbers are mapped sequentially to DYSE channel numbers, which define the external connections to the emulator, and are written to a node mapping text file to be used by the operator. Channel impulse response output files are generated for each transceiver pair and point in time of the study. The DYSE scenario file consists of line entries made up of the following per-link information: timestamp, transmitter number, receiver number, Doppler shift, gain, phase, and delay for three paths. The full CIR data is extracted for all transceiver pairs at each time step for processing. Within a time step, the Doppler shift is calculated for each transceiver pair based on velocity values from the configuration file. The three strongest paths between transceiver pairs are also extracted and combined with the remaining line entry information prior to being written to file. Directional gains from antenna pattern data may be applied optionally at a transceiver node in post-processing if desired. The extracted DoA or DoD angles corresponding to the CIR

path data are interpolated using the angular gains and applied on a per-link basis prior to tap reduction. Antenna patterns may also be defined in the Wireless Insite project itself; however, implementation in post-processing reduces the overall computation time required if multiple patterns are considered, such as the case for reconfigurable antennas. An overview of the aforementioned Wireless Insite post-processing to the DYSE format procedure is shown in Figure 8.

A node mapping file is used by the operator to correctly connect external devices under test to the DYSE channel numbers, which match the generated scenario file. The maximum input power permitted by DYSE is -20 dBm, which often requires additional attenuation to be applied at the input port. A JSON configuration file is used to define the following parameters for each DYSE channel used: (i) a fixed attenuation measured from a spectrum analyser, (ii) additional transmit power, and (iii) the option to re-map to a different DYSE channel number if desired. The dynamic spectrum environment emulator DYSE channel numbers may be re-mapped to other channels to avoid the tear-down of other experimental setups connected physically to the emulator. In addition, the ability to extend the scenario through repetition in time is an optional parameter. The dynamic spectrum environment emulator configuration files specifying emulation time, channel

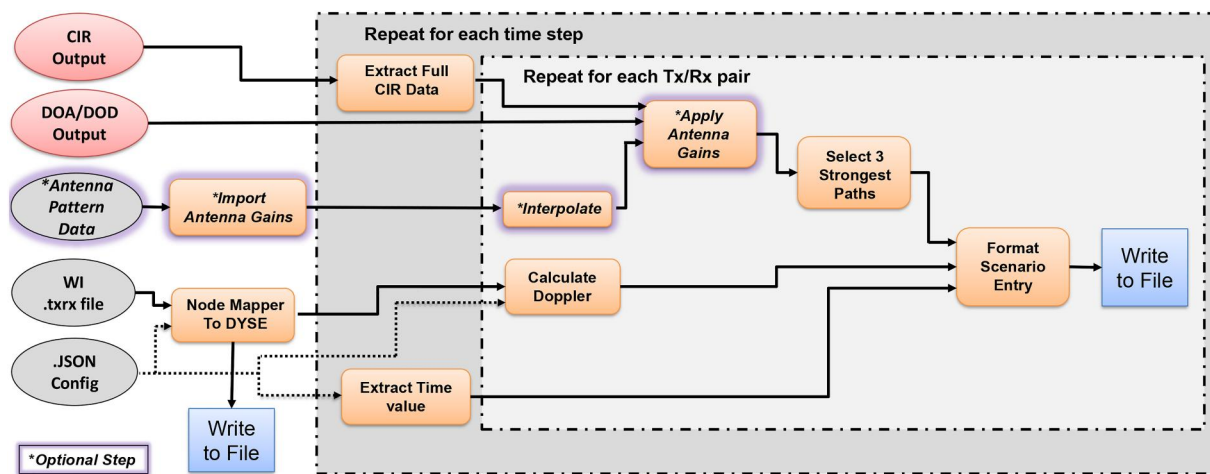


FIGURE 8 Wireless Insite post-processing for dynamic spectrum environment emulator (DYSE)

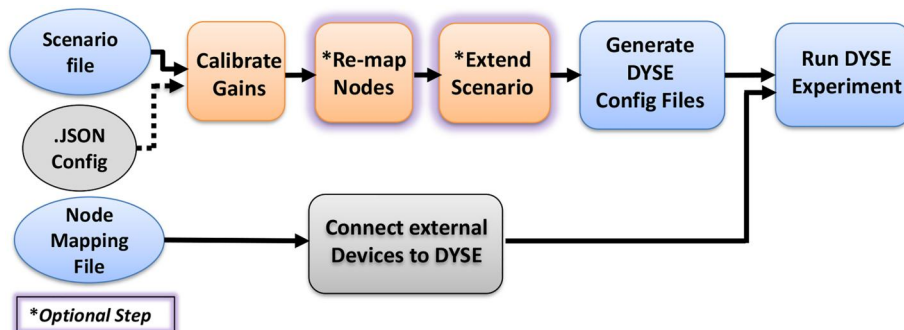


FIGURE 9 Dynamic spectrum environment emulator (DYSE) pre-processing and execution

update rate, carrier frequency, and channels in use are automatically generated based on the input scenario file and JSON options selected. The DYSE emulator is run in batch mode using the aforementioned configuration options and the resulting scenario file, with underlying C and C++ code used to control the emulator via socket programming. DYSE pre-processing and execution is summarised in Figure 9.

4 | CASE STUDIES

Cyber-Physical Systems include UAVs [35], vehicular technologies, and critical infrastructure control and defence [36]. In this section, we analyse the specific examples of using the experimentation framework for (i) mitigating the effects of a constant jammer during UAV communication through the use of electrically reconfigurable antennas, (ii) adjusting time scheduling of vehicle to infrastructure communication systems under periodic and random jamming, and (iii) designing a reactive jammer to defend infrastructure systems against an illegally operating rogue IoT network. The flexibility of our framework enables the creation of various new scenarios with different dynamic components, physical environment, antenna, and waveform attributes.

4.1 | UAV communication

Unmanned aerial vehicles have the potential to solve many upcoming challenges in civil, industrial, and military domains due to flight speed capabilities, high manoeuvrability, and low-cost implementation. Applications may consist of disaster relief, data collection, hot-spot base stations, and countless others. The unique 3-dimensional nature of UAV communication leads to difficulties related to environmental factors, antenna radiation patterns, varying distances, shadowing from the physical UAV, interference, and jamming. Additional work needs to be done at the PHY layer to overcome these challenges [37]. Similarly to the eavesdropping mitigation techniques proposed for IoT networks, friendly jamming with UAV communication systems has been considered to compensate for the susceptibility of eavesdropping attacks [38–41]. Air-to-ground channels are predominately line-of-sight (LOS) and lead to open exposure to unintended parties. Mitigation techniques to protect against a UAV-based jammer have also been provided in [4]. UAVs have been proposed as relay nodes to assist in vehicular communications with roadside infrastructure and edge devices in efforts to mitigate jamming attacks [5, 6]. In all of the aforementioned studies, the only form of the evaluation has been through simulation. More realistic testing of UAV communication scenarios with jamming must be performed to assess the suitability of the techniques. An SDR-based jammer was developed to target UAV control signals in [26]; however, very limited experimentation was performed, as the authors only considered a simple wired connection. Overall, a method must be developed to realistically test emerging wireless technologies under complex jamming scenarios.

4.1.1 | Ray-tracing scenario

We consider the case in which a low-mobility UAV is performing as an access point to 4 ground users, under the presence of a constant jammer. The ground users use electrically reconfigurable antennas [42] to mitigate the effects of the jammer. The different antenna patterns are applied as filters to the result of the ray-tracing study for the purpose of reducing computation time. Each transceiver uses an omni-directional vertical antenna at a 3-dBi gain with a carrier frequency of 2.4 GHz. The study area is based on the campus of Drexel University in Philadelphia, consisting of student walkways, campus buildings, and parks, as shown by the Google Earth capture in Figure 10. The UAV travels at a rate of 0.5 m/s with an altitude varying between 6 and 7 m over a circular trajectory with a radius of 7 m. The ground nodes use antennas with heights of 2 m and travel randomly within a bounded area of approximately 50 by 50 m at a rate of 0.25 m/s. The jammer is located approximately 32 m from the centre of the trajectory and also has an antenna height of 2 m. A model of the selected area is developed for Wireless Insite in [31]. The scenario features are shown in Figure 11, with concrete buildings in tan colour, grass parks in green, and concrete walkways in grey. The UAV travels over the light blue dotted trajectory shown and the ground nodes are represented by blue triangles and travel within the black dotted boundary while the constant jammer is shown as a red circle. The ground nodes maintain a minimum distance of 20 m from one another and each independently changes travel direction randomly every 15 s. The total run-time of the study is 15 min, which corresponds to the UAV completing the trajectory 3 times. As mentioned previously, the reconfigurable antenna patterns are applied outside of the ray-tracing calculations. The planar reconfigurable

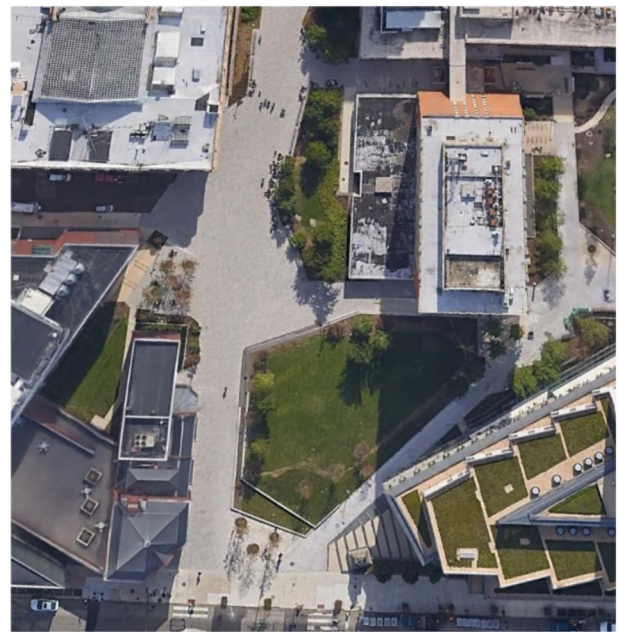


FIGURE 10 Google Earth capture of Drexel area. Focused on south-side of 3200 market block in Philadelphia

antenna described in [42], known as the Reconfigurable Alford Loop Antenna (RALA), was selected. The RALA operates at 2.4 GHz and is capable of supporting omni-directional, 2

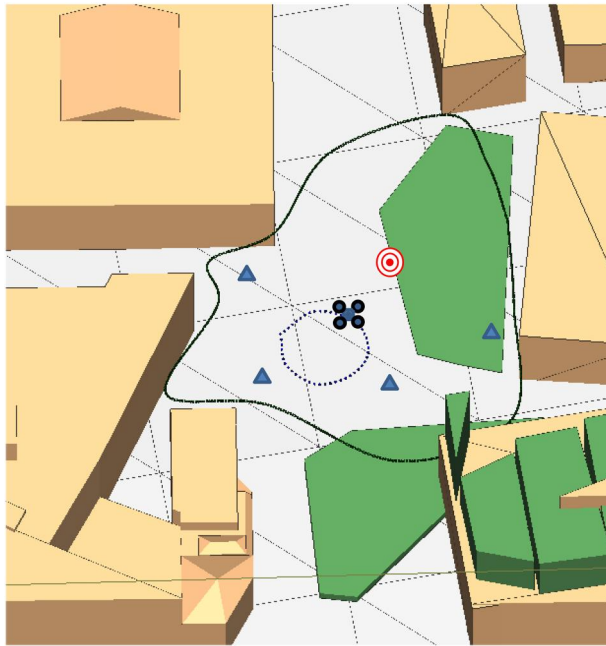


FIGURE 11 Wireless Insite model of Drexel area. UAV trajectory shown in the light blue line, initial ground user location in blue triangles, bounding area for ground user movement shown in the black line, and constant jammer by the red circle

bi-directional, and 4 single-directional beams. In our study, we consider the omni-directional and 4 directional states of the antenna. RF channels are evaluated between each of the ground nodes, the UAV access point, and the jammer using the baseline omni-directional antennas in Wireless Insite. The channels involving the ground nodes are updated to reflect the angle of arrival (AoA) or angle of departure (AoD) of the received or transmitted signals, respectively. Path loss is adjusted per link based on the phi and theta antenna gains of each RALA state over the entire duration of the study. The resulting output consists of scenarios for the baseline omni-directional antenna, the omni-directional state of the RALA for the ground nodes, and for each of the 4 directional states. A summary of the channel emulation parameters used for the aforementioned scenarios is provided in Table 3. Received power statistics, mean delay spread, and Doppler shift are shown for the conventional omni-directional antenna and different states of the RALA from the ground nodes receiving the Jammer and UAV signals, respectively. The listed values are based on the links from all ground nodes in the experiment with the jammer transmit power of 10 dBm and the UAV transmit power of 20 dBm. In addition, the link between the UAV and constant jammer is provided, in which both use the conventional omni-directional antennas. The mean received power from multi-path components MP1 and MP2 are shown relative to the strength of the LOS component. Doppler shift of 2 Hz is applied based on the 2.4 GHz carrier frequency and velocities of the nodes. Comparisons of the mean LOS receive power between the Jamming and UAV transmit signals, as shown in bold font, demonstrate that there is an overall benefit

TABLE 3 Channel emulation parameters for unmanned aerial vehicle (UAV) communication experiment

Rx	State	Tx	LOS Rx power (dBm)				Mean Rx power (dBm)		Mean delay Spread (μ s)	Doppler shift (Hz)
			Min	Max	Mean	STD	MP1 (relative)	MP2 (relative)		
UAV-OMNI	N/A	Jammer	-57.92	-53.59	-56	1.48	-15.3	-18.45	1.75	4
GN-OMNI	N/A	Jammer	-61.06	-45.71	-55.22	4.15	-9.47	-12.49	1.04	2
		UAV	-55.48	-37.01	-43.49	2.69	-16.37	-19.89	1.81	2
GN-RALA	0	Jammer	-60.76	-45.34	-54.51	4.3	-9.58	-12.36	1.03	2
		UAV	-64.26	-36.67	-42.83	2.61	-16.29	-19.76	1.81	2
GN-RALA	1	Jammer	-74.28	-45.26	-58.28	6.29	-5.46	-10.99	1.1	2
		UAV	-72.09	-35.74	-48	8.1	-15.34	-18.16	1.92	2
GN-RALA	2	Jammer	-60.34	-46.1	-54.91	4.16	-10.15	-16.18	1.02	2
		UAV	-66.66	-35.46	-50.15	7.61	-17.99	-21.25	2.02	2
GN-RALA	3	Jammer	-79	-44.82	-63.68	8.55	-9.95	-13.9	1.31	2
		UAV	-64.37	-35.22	-48.73	6.56	-16	-20.72	1.95	2
GN-RALA	4	Jammer	-78.58	-49.84	-67.56	6.07	-4.28	-5.17	1.58	2
		UAV	-60.29	-34.59	-46.78	5.93	-14.1	-18.09	1.89	2

Note: Summary of channel properties used for links between conventional omni-directional antennas at UAV and ground nodes, RALA receive antenna at ground nodes, constant jammer using the omni-directional antenna with 10-dBm transmit power and UAV using the omni-directional antenna with 20-dBm transmit power. Different antenna states are shown for the RALA antenna. Antenna states 3 and 4 reduce the effects of the jammer relative to the UAV link in comparison to the conventional omni-directional antenna as shown in bold. Abbreviations: RALA, reconfigurable Alford loop antenna; UAV, unmanned aerial vehicle.

of using RALA states 3 and 4 in comparison to the conventional omni-directional antenna. The effects of the jamming signal are decreased, relative to the received signal from the UAV.

4.1.2 | Experimental setup

The hardware infrastructure described in [19] is leveraged to provide the devices under test and active threat connected to the DYSE channel emulator. Six SDR grid nodes, each using an Ettus USRP N210 radio, are configured to use the wired connection to the DYSE path panel. The primary network consists of five DragonRadio nodes using OFDM at a carrier frequency of 2.4 GHz, signal bandwidth and sampling rate of 5 MHz, and TDMA operation. A time slot size of 50 ms with a guard interval of 5 ms is selected. The default DragonRadio settings of 48 sub-carriers, cyclic prefix length of 6, adaptive modulation and coding, Automatic Repeat Request, and selective acknowledgements are used. A summary of the DragonRadio network configuration is provided in Table 4. One of the DragonRadio nodes is selected to be the UAV access point and is allocated 60% of the TDMA time slots; each of the four ground nodes is allocated 10% of the time slots. The UAV transmits at a power of 20 dBm while the ground nodes transmit at 13 dBm. The remaining SDR grid node is set up as a constant jammer using the UHD Signal Generator from GNU Radio [43]. The jammer transmits a Gaussian waveform at 2.4 GHz with a signal bandwidth of 5 MHz to directly match the DragonRadio network. The Multi-Generator (MGEN) Network Test Tool [44] is used to transmit concurrent 820 kbps periodic User Datagram Protocol (UDP) data links to each of the ground nodes from the UAV. Each ground node writes the received MGEN data to a log file, which is then used for the performance analysis. The study is repeated for each of the scenarios described in Section 4.1.1, with varying jammer signal strength. The DYSE scenarios consist of the ground nodes using (i) the generic omni-directional antennas; (ii) each of the single-directional states of the RALA; and (iii) the omni-directional state of the RALA. The jammer transmits at 10 dBm, corresponding to 3 dB weaker than the power used by the ground nodes. The baseline case in which no jammer is present in the study is also considered. Overall a total of 12 situations are examined in this experiment: the 6 unique channel emulator scenarios, each with and without the constant jammer.

4.1.3 | Performance evaluation

MGEN data flows are each tagged with sequence numbers and timestamps to allow for the performance to be analysed in post-processing. The Tcpdump Rate Plot Real Time (TRPR) tool parses MGEN log files to determine various metrics such as data rate, latency, and loss [45]. Log files were collected at each ground node receiver, for all channel emulator scenarios based on antenna selection and different jamming strengths.

TABLE 4 DragonRadio configuration for unmanned aerial vehicle (UAV) communication study

PHY	OFDM	MAC	TDMA
Carrier frequency	2.4 GHz	Signal bandwidth	5 MHz
No. of nodes	5	Sampling rate	5 MHz
No. of sub-carriers	48	Cyclic prefix Ln.	6
Slot size	50 ms	Guard size	5 ms

Abbreviations: MAC, Medium Access Control; OFDM, Orthogonal Frequency Division Multiplexing; TDMA, Time-division multiple access.

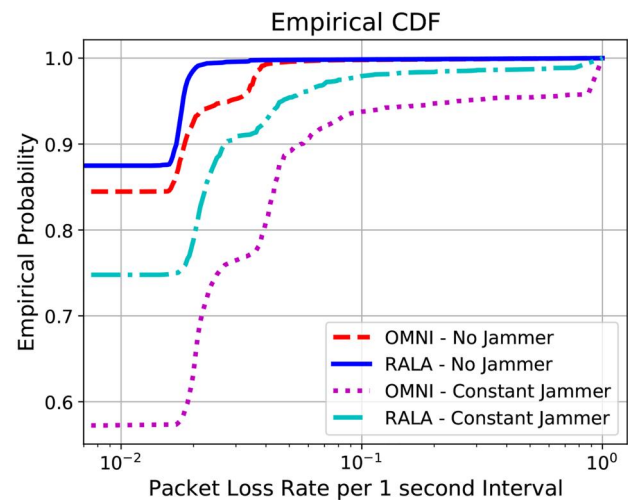


FIGURE 12 Empirical cumulative distribution function (ECDF) of a packet loss rate for unmanned aerial vehicle (UAV) study. Comparison of the conventional omni-directional antenna and reconfigurable Alford loop antenna (RALA) with and without a constant jammer. The empirical cumulative distribution function consists of all receiving nodes and the loss rate is calculated over 1 s intervals

The performance metric of interest, packet loss rate, is calculated in TRPR based on received sequence numbers within an averaging window. Tcpdump rate plot real time returns average metric values at the start and end of each window. We consider the maximum loss rate calculated during a 1 s averaging interval. Throughput is also calculated during matching time windows and is used to set the loss rate to 1 if the throughput does not exceed a threshold of 10 kbps in the corresponding interval. The RALA state of each ground node is determined by selecting the minimum loss rate at each 1 s interval resulting from the RALA omni-directional and four directional mode log files. The movement of the UAV and ground nodes during the scenario cause performance to be time-varying and different for each node. We calculate the Empirical Cumulative Distribution Function (ECDF) of the packet loss rate metric to demonstrate the overall system performance for all receiving nodes. Figure 12 shows a comparison of the ECDF of the packet loss rate for the omni-directional antenna and RALA under normal operation and with the constant jammer. Without the jammer present, the empirical probability to achieve a loss rate of 0.01 is 0.88 for the RALA case in

comparison to the conventional omni-directional antenna with a probability of 0.855. The results, in general, highlight that the improved receive gain from the directionality of the RALA enhances system performance in comparison to an omni-directional antenna. Under the presence of the constant jammer, the empirical results show that during 90% of the experiment the conventional omni-directional antenna is capable of maintaining a loss rate of 0.055, whereas the use of the RALA improves the loss rate to 0.025 during the same period. The results demonstrate that the use of reconfigurable antennas can help improve the performance of wireless systems under the presence of an RF jammer. An important consideration is that the reconfigurable antennas were only used at the ground receivers. Using reconfigurable antennas at both ends of the link should further improve performance. Overall the comparison between the use of a standard omni-directional antenna and a RALA at the ground nodes demonstrates that reconfigurable antennas effectively mitigate the effects of a jammer. Future work involves the real-time selection of RALA antenna states by the nodes based on learning techniques.

4.2 | Vehicular technologies

Emerging Cyber-physical systems focus on intelligent and interconnected devices. The Dedicated Short-Range Communications (DSRC) frequency band of 5.9 GHz has been designated for vehicular communication. Wireless Access in Vehicular Environments (WAVE) is defined by the IEEE 802.11p standard, using OFDM signals with 10 MHz of bandwidth at the DSRC band [46, 47]. Inter-vehicle and vehicle to roadside infrastructure communication are vital components of WAVE with the primary goal of reducing collisions between automobiles [48]. Communication systems based on DSRC must be robust and well protected to mitigate collateral damage and more importantly help protect human lives. Signal jamming has been shown to be an issue for WAVE and other communication systems within automobiles in the past [49, 50]. Vehicle communication systems must be resistant to external security threats such as denial of service and false data injection. Jammers are capable of sending white Gaussian noise or false data over the medium. Additional work must be performed on reactive security and context verification, in which a vehicle can realise it is being attacked by comparing received information related to status and environment with its own information it has [51]. The communication system should determine if it is being tampered with independent of the application, as protection at the PHY layer will improve the reliability of WAVE. The simulation framework in [7] was used to analyse car to infrastructure scenarios under the presence of a constant jammer. Performance results demonstrated correlation with experimentally collected data over identical GPS coordinates; however, the environment was run entirely on a host machine, was limited to the PHY layer, and only considered the free-space path loss model. More complex test cases must be analysed with hardware-in-the-loop, full-stack

radios such as we describe in Section 3.2.1, and accurate channel models. Recent work has shown a benefit in using directional antennas in comparison to omni-directional antennas primarily due to the ability to focus radiation under tunnel scenarios [52–54]. Omni-directional antennas in the horizontal plane with low vertical beamwidth have also been shown to be suitable for vehicular communications [55, 56].

4.2.1 | Ray-tracing scenario

In this scenario, a Honda Civic with a DSRC-like receiver attached to the centre of its roof is driving over the Benjamin Franklin Bridge, which connects Philadelphia (Pennsylvania) to Camden (New Jersey), while attempting to maintain communication with infrastructure units spread throughout the vehicle's path. Jammers capable of using periodic and random attacks are co-located with the infrastructure units. The legitimate system uses adjusted time scheduling in an effort to mitigate the effects of the jammers.

A Google Earth capture of the bridge is shown in Figure 13. A Wireless Insite model was developed consisting of different bridge materials such as metal, concrete, brick, and asphalt for the road based on the Sketchup model of [57]. In the immediate area, there is a water body representing the Delaware River. There are 10 total traffic lights located on the bridge with heights approximately 8 m above the road. Along the bridge, we consider six infrastructure units grouped into 3 pairs of nearby traffic lights, with each node located 1 m below the lights. At each infrastructure pair, there is a DragonJammer located equidistant from the nodes, 1.5 m above the concrete walkway, which is elevated from the road. The first pair is located at traffic light numbers 2 and 3, relative to heading in the east direction, with a DragonJammer separated by approximately 138 m from each. In the middle of the bridge, there are infrastructure nodes on traffic lights 4 and 6, with a jammer located roughly 265 m in between the two. The third pair is on traffic lights 7 and 8 on the east portion of the bridge, with a DragonJammer located approximately 127 m from each node. In our study, all nodes (car,



FIGURE 13 Google Earth capture of the Ben Franklin bridge. Bridge connects Philadelphia, PA to Camden, NJ. Above shows sky-view of the entire bridge, side-profile of bridge centre below

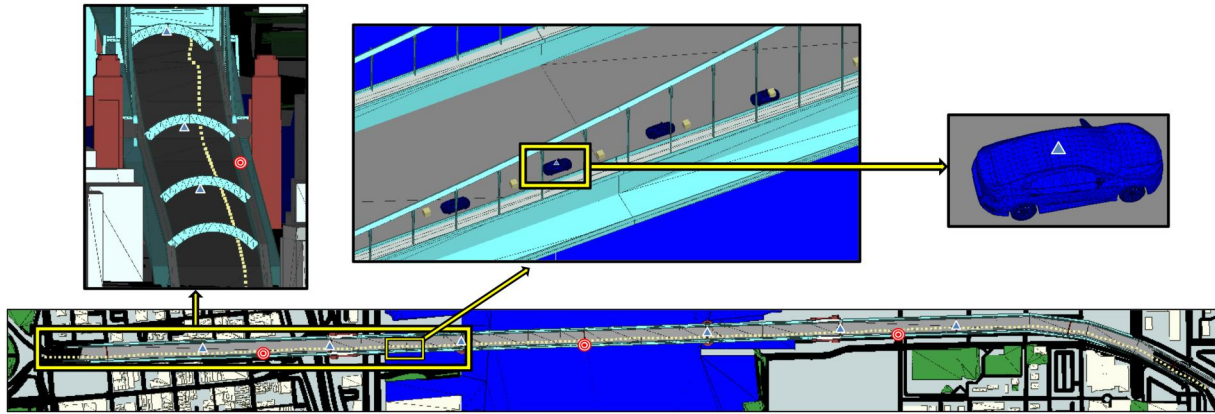


FIGURE 14 Wireless Insite model of the Ben Franklin bridge. Dedicated short-range communication transceivers for infrastructure and automobile shown by blue triangles, jammers by red circles, automobile trajectory by the yellow dashed line

infrastructure, and jammers) use the vertically polarised omnidirectional antenna with a 14° vertical beamwidth and a gain of 9 dBi, described in [55]. The narrow vertical beamwidth of the antennas results in null zones directly underneath the transceivers; therefore, the infrastructure nodes were placed in pairs to provide continued coverage. A convoy of 4 Honda Civic cars, each separated by approximately 18 m, travels at a rate of 17.88 m/s over the duration of the bridge for a total distance of approximately 2.5 km. The cars are based on the Sketchup model in Reference [58]. The third car of the convoy is equipped with a receiver located 0.1 m above the centre of the exterior roof, which has an approximate height of 1.5 m. The ray-tracing study is performed at a carrier frequency of 5.9 GHz to represent DSRC communications and is updated at a rate of once per second during the travel of the cars from the west-side to the east-side of the bridge. An overview of the described ray-tracing scenario is shown in Figure 14. Infrastructure and car nodes are represented by blue triangles, DragonJammer nodes by red circles, and the trajectory of the cars by the dashed yellow line. Metal objects are shown in light blue for the bridge and dark blue for the cars, asphalt in dark grey, concrete walkways in light grey, brick in red, water bodies in dark blue, and grass parks in green. A focus on the first infrastructure pair at the second and third traffic lights is shown in the upper left, the convoy is highlighted in the centre, and a closer viewpoint of the Honda Civic equipped with the receiver is shown on the right. A summary of the channel emulation parameters used in the scenario is provided in Table 5. Receive power statistics of the LOS component are shown for links between the vehicle, infrastructure roadside units RSU, and the DragonJammer nodes. All devices transmit at 33 dBm. Calculated values are based on the RSU and DragonJammer nodes, which provide the strongest receive power to the vehicle at each point in time during the study. Multi-path components are negligible in comparison to the LOS path due to the high directionality of the antennas selected. The mean delay spread observed is on the order of nanoseconds for instances in which a multi-path component does exist. A Doppler shift of 351.67 Hz is applied based on the 5.9 GHz carrier frequency and vehicle speed of 17.88 m/s.

TABLE 5 Channel emulation parameters for vehicular experiment

Rx	Tx	LOS Rx power (dBm)				Doppler Shift (Hz)
		Min	Max	Mean	STD	
Vehicle	RSU	-52.48	-23.47	-42.07	5.71	351.67
Vehicle	Jammer	-70.38	-36.08	-46.75	6.73	351.67
RSU	Jammer	-56.66	-39.15	-42.51	4.96	0

Note: Summary of channel properties used for links between vehicle, infrastructure roadside unit nodes RSU, and DragonJammer nodes. All devices use 33-dBm transmit power. Values are based on the strongest line-of-sight (LOS) path from RSU and jammer to the vehicle at each point in time.

4.2.2 | Experimental setup

The grid SDR network testbed described in [19] has been expanded to include additional Ettus USRP X310 radios; 8 of these units were used for this experiment. A 7 node DragonRadio network is set up for the vehicle and infrastructure devices, and one node is used as a DragonJammer. The DragonJammer signal is split into 3 parts and fed into separate input channels for the emulator, such that each of the jamming signals described in Section 4.2.1 is identical yet undergo different channel effects. DragonRadio is configured to use the TDMA-FDMA MAC, a signal bandwidth of 10 MHz, and a sampling rate of 20 MHz. The remaining settings match the parameters described in Section 4.1, and a summary of the DragonRadio attributes is provided in Table 6. The carrier frequency of the radios in this experiment is set to 4.2 GHz during operation to match the input range of the RF splitters used for the jamming signals. An important note is that since the ray tracing was performed at 5.9 GHz, the channels experienced between the devices are representative of vehicular communication, despite the lower frequency used by the physical radios. The TDMA/FDMA MAC provides flexibility to DragonRadio nodes in both time and frequency domains, such that nodes can be allocated different channels and time slots. Each node is allocated its own 1-MHz channel in which further scheduling may be performed by assigning time slots.

TABLE 6 DragonRadio configuration for vehicular communication study

PHY	OFDM	MAC	TDMA/FDMA
Carrier frequency	4.2 GHz	Signal bandwidth	10 MHz
No. of nodes	7	Sampling rate	20 MHz
No. of sub-carriers	48	Cyclic prefix Ln.	6
Slot size	50 ms	Guard size	5 ms

Note: Ray tracing is performed at a carrier frequency of 5.9 GHz.

Abbreviations: FDMA, Frequency-division multiple access; MAC, Medium Access Control; OFDM, Orthogonal Frequency Division Multiplexing; PHY, Physical; TDMA, Time-division multiple access.

MGEN is used to transmit concurrent 820 kbps periodic UDP data links from each infrastructure node to the vehicle transceiver node. We consider the default schedule *S0* with nodes assigned all time slots within their dedicated channels, and a modified schedule *S1* in which the first quarter of the 1 s frame is not used for transmission. DragonJammer has 2 modes in this experiment. The periodic mode transmits a chirp signal with a duration of 0.25 s and rest of 0.75 s, whereas the random mode transmits a chirp signal for a random duration between 0.1 and 1 s and rests for the complement of the transmission time. We use a transmit power of 33 dBm for each node, including the jammers, to match the maximum transmit power of DSRC systems [59]. We examine the following situations using the ray-traced channels: (i) default schedule *S0* with no jammer present, (ii) default schedule *S0* with a periodic jammer, (iii) modified schedule *S1* with a periodic jammer, and (iv) modified schedule *S1* with a random jammer.

4.2.3 | Performance evaluation

Multi-generator network test tool log files are collected at the vehicle receiver for each different jamming and scheduling scenario. Tcpdump rate plot time is used to provide packet performance metrics for all data links sent by the infrastructure nodes, following the approach described previously in Section 4.1.3. The maximum packet loss rate within a 0.5-s window is taken on a per-link basis. The loss rate is then set to 1 at a particular observation window if the link does not maintain a 250 kbps throughput within the averaging period. The receiver log file contains information for each of the links sent by the infrastructure nodes. We consider the vehicle to be maintaining constant communication with roadside units during its travel across the bridge; therefore, the packet loss rate at the vehicle is taken as the minimum resulting value calculated from all links within each 0.5-s window. The ECDF of the packet loss rate is analysed for the different jamming and scheduling configurations in Figure 15. The baseline case of default schedule *S0* and no jammer present is shown in the blue dash-dot line, demonstrating that loss occurs minimally when there is no active threat. The empirical results of schedule *S0* under the periodic jammer, represented by the red dashed trace, illustrate a significant impact

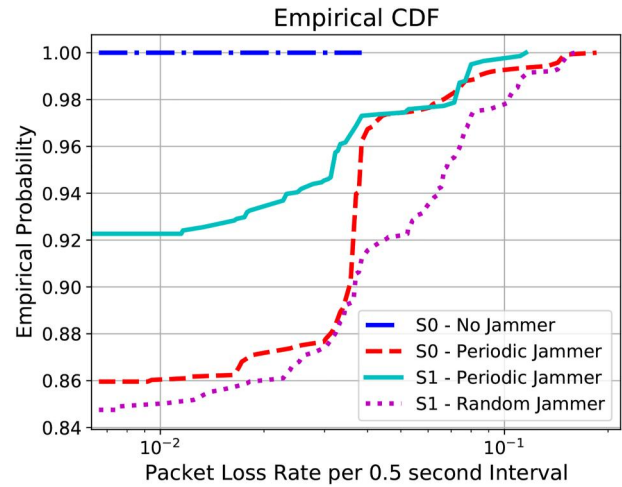


FIGURE 15 Empirical cumulative distribution function (ECDF) of the packet loss rate for vehicle study. Comparison between different jamming types and DragonRadio medium access control (MAC) schedules. Schedule *S0* follows the default configuration, *S1* is modified to mitigate periodic jammer. The loss rate is calculated over 0.5 s intervals

on the performance with a maximum error rate of 0.01 occurring 86% of the time. Modifying the schedule to *S1* increases the percentage of the experiment that the maximum error rate was detected to 92%, as shown by the cyan solid line. The higher empirical probability at the same packet loss rate value demonstrates that the system is able to maintain a better communication link by using a modified timing schedule. The results motivate the development of adaptive scheduling techniques based on observations and learning to mitigate the effects of jamming. The ECDF of the modified schedule *S1* under the random jammer is shown by the magenta dotted trace, with an empirical probability of 0.85 at the 0.01 packet loss rate. The performance exemplifies that the modified timing schedule does not mitigate the random jammer and inspires future work in random scheduling and frequency-agile methods.

4.3 | Defence infrastructure

Reactive jammers may be used to improve the quality of licensed devices in the presence of illegitimate systems by specifically targeting the threats [24, 60]. Internet-of-Things devices have traits of limited signal processing, limited storage/memory, and compact form factors. Long-range applications use technologies such as LoRa for wide-area networks; however, short-range IoT systems focus on Wi-Fi, Bluetooth, and ZigBee technologies. Healthcare [9] and consumer IoT networks [61] have both been shown to be vulnerable to jamming attacks in the past. An additional issue exists of premature battery draining of devices resulting from re-transmissions during active jamming.

Intentional jamming has been examined previously for IoT systems to mitigate eavesdropping threats; however, limited experimentation has been performed to evaluate the proposed designs and energy efficiency trade-offs [8, 38, 62].

Defensive intentional jamming methods can be improved with a further evaluation under different scenarios with practical constraints of physical components considered.

4.3.1 | Ray-tracing scenario

We assume the case of a rogue IoT Wi-Fi-based network operating on a restricted carrier frequency. The network consists of four station nodes (STA), which communicate sensor data with 1 central access point (AP). A reactive jammer is implemented as a defence mechanism to disrupt the rogue IoT network, without rendering the frequency band completely useless. A simple jammer that does not transmit based on detection may render the medium useless for licensed systems. In particular, we consider a rogue network to be operating on Wi-Fi channel 12 (2.467 GHz) outside the Philadelphia Museum of Art. A Google Earth capture of the museum is shown in Figure 16. The STA nodes move randomly within a bounded area along the entrance of the building, at a velocity of 0.5 m/s, while the AP remains stationary in the centre of the platform. Each node is 1.6 m off of the ground, and the bounded area is approximately 60 m². The STA nodes change travel direction every 5 s. A reactive jammer is placed on the edge of the front of the museum roof at a height of 40 m above the rogue network nodes. The jammer is capable of transmitting signals with variable duration and signal strength. A Wireless Insite model was developed for the aforementioned scenario as shown in Figure 17 based on the Sketchup model in [63]. The reactive jammer is shown by a red circle, AP by a blue and white diamond, initial STA locations by blue triangles, and the bounding area for STA nodes is represented by a carmine



FIGURE 16 Google Earth capture of the Philadelphia museum of art. Front-facing view shown

dotted line. Each transceiver in the study uses an omnidirectional vertical antenna at a 3-dBi gain. Concrete materials are shown in light grey, brick in yellow, metal in dark grey, and dry Earth in green. The total scenario time is 5 min long, with channel updates made once per second. A summary of the channel emulation parameters is provided in Table 7. Receiver power statistics are shown for the dominant component of links between the AP, reactive jammer (RJ), and each of the 4 STA nodes. Multi-path received power is provided relative to the LOS component and associated mean delay spread. A Doppler shift of 4.11 Hz is applied for the STA nodes based on the 2.467 GHz carrier frequency and 0.5 m/s velocity. The STA-2 node becomes briefly out of range for both the jammer and AP when it turns the front corner of the building, as reflected by the minimum main path Rx power and high standard deviation. The node located initially on the stairs, STA-4, has a poorer channel with the AP as expected due to the lack of a LOS path. In contrast, the height profile of the reactive jammer ensures a strong channel with STA-4.

4.3.2 | Experimental setup

The rogue IoT network is set up using Wireless Open Access Research Platform (WARP) v3 SDRs, devices external from the SDR Grid Testbed, developed by Mango Communications, Inc. [64]. We use the 802.11 Reference Design with 1 node configured as an access point and 4 nodes as stations. Each radio is cabled into DYSE using RF circulators and

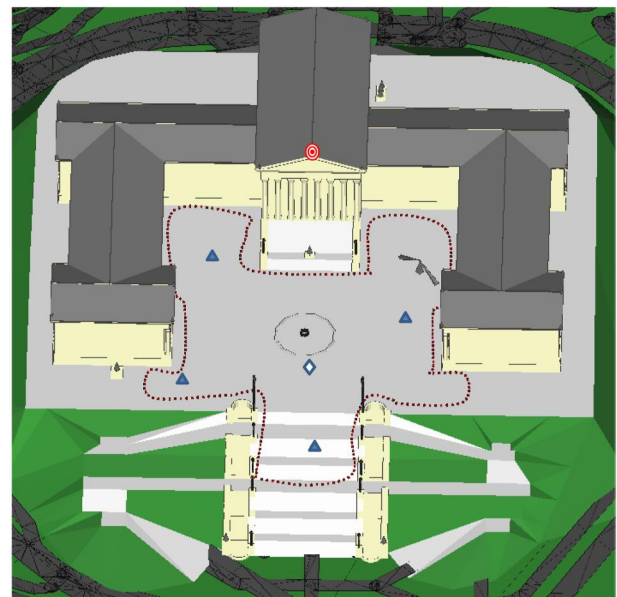


FIGURE 17 Wireless Insite model of the Philadelphia museum of art. Reactive Jammer shown by the red circle, a stationary central node in blue and white diamond, initial moving node locations by blue triangles, and bounding area for nodes shown in a carmine dotted line

TABLE 7 Channel emulation parameters for defence infrastructure experiment

Rx	Tx	Main path Rx power (dBm)				Mean Rx power (dBm)		Mean delay Spread (μ s)	Doppler shift (Hz)
		Min	Max	Mean	STD	MP1 (relative)	MP2 (relative)		
AP	RJ	-63.02	-62.10	-62.37	0.20	-14.74	-16.98	1.19	0.00
STA-1	RJ	-63.87	-57.47	-60.03	1.70	-14.33	-17.87	1.78	4.11
	AP	-60.43	-56.14	-58.50	1.14	-9.91	-10.63	1.76	4.11
STA-2	RJ	-137.00	-60.42	-62.66	8.68	-18.10	-24.17	0.71	4.11
	AP	-137.00	-56.19	-58.25	9.18	-14.95	-18.65	2.80	4.11
STA-3	RJ	-58.95	-58.06	-58.35	0.15	-13.56	-23.05	1.17	4.11
	AP	-58.85	-55.24	-57.40	0.81	-11.54	-13.85	3.16	4.11
STA-4	RJ	-91.49	-63.25	-64.75	3.48	-14.71	-18.88	1.14	4.11
	AP	-126.25	-45.31	-84.18	25.35	-5.61	-8.49	1.94	4.11

Note: Summary of channel properties used for links between rogue IoT access point (AP), stations (STA), and reactive jammer (RJ) using equal transmit power. Omni-directional antennas are used by all nodes with 13-dBm transmit power. Channels are reciprocal. Mean received powers of multi-path components MP1 and MP2 are shown relative to the received power from the dominant line-of-sight path.

external attenuation at the transmitter port. The Python experiments framework is leveraged with WARP boards connected through an Ethernet switch to a single computer. Wi-Fi channel 11 (2.462 GHz) is used to avoid modification of the basic design, although the ray tracing described in the previous section was performed at channel 12 (2.467 GHz). We use the non-high throughput OFDM with 16-QAM half coding rate configuration with a target throughput of 24 Mbps for the rogue network. The distributed coordination function MAC is used with 3 maximum transmissions and a beacon signal is sent every 10 ms. Each station has a traffic flow sent to the access point using a local generator with a payload length of 1400 bytes. The WARP nodes use a transmit power of 13 dBm. Identically sized links are sent from the access point to each station node. We use the experiment framework to calculate the received throughput at each link for 1 s intervals. The reactive jammer described in Section 3.2.2 is launched on an Ettus N210 radio connected to the SDR Grid Testbed. The node is set to a carrier frequency of 2.462 GHz and uses a receive gain of 20 (the USRP middle-point). The matched filter compares the received signal to the IEEE 802.11 short training sequence, which is used for synchronisation. An AWGN signal is transmitted without any delay and for variable transmit gain and duration. We consider transmit powers equal to the rogue IoT nodes, as well as 3 dB weaker and 3 dB stronger for comparison. Duration is examined in 50 μ s increments from 50 to 3000 μ s. In addition, smaller durations of 5 and 10 μ s are also assessed. The 5-min scenario, described in the previous section, is repeated for each duration and transmit power configuration of the reactive jammer through the channel emulator. A baseline case without the jammer present is also considered. Log files are collected for the throughput performance of the rogue IoT network with each of the described jamming configurations.

4.3.3 | Performance evaluation

The aggregate throughput for all communication links is taken at every 1-s observation window for all of the collected log files from the experiment. The median, mean, and standard deviation are calculated for the total run of each jamming configuration. A summary of the aggregate throughput statistics of the rogue IoT network is provided in Table 8. The baseline performance of the rogue IoT network without a jammer present is provided at a duration of 0 μ s and is shown to meet the expected throughput of 24 Mbps. We noticed that the mean aggregate throughput eventually plateaued despite increasing jamming duration due to a small number of observation windows in which the rogue IoT network is able to achieve higher throughput in comparison to the entire emulation time. We have determined that the median, rather than the mean, is a more accurate statistic to consider for system performance. The expected trends were observed of decreased aggregate throughput with increased jamming duration or transmit power. We did not examine jamming duration larger than 500 μ s due to a plateau in performance. Based on our results from Table 8, we determined a 13-dBm transmit signal with 50 μ s duration to be a preferred setting for defence against the rogue IoT network. A reactive jammer, in comparison to a simple jammer, allows for the targeting of specific signal types and is capable of performing in a power-efficient manner. The rogue IoT network is disrupted by the reactive jammer, which may enable licenced systems to communicate over the same frequency band. Power efficiency, as represented by lower transmit power and duration, allows for devices to run for longer periods of time if operating on a battery. Shorter jamming duration and lower transmit power are also helpful for opening the RF medium for legitimate communication systems. In summary, we have observed that a reactive jammer may be configured, with a consideration of

TABLE 8 Aggregate throughput (Mbps) of rogue IoT network with reactive jammer. Statistics shown for different jamming duration and power

Dur. (μ s)	10 dBm			13 dBm			16 dBm		
	Med.	Mean	STD	Med.	Mean	STD	Med.	Mean	STD
0	23.13	24.68	2.81	-	-	-	-	-	-
5	22.62	23.68	2.49	16.93	16.18	3.90	11.31	11.59	2.97
10	13.37	12.26	3.26	2.97	3.73	2.99	0.08	1.33	2.60
50	7.31	7.12	4.00	0.01	1.31	2.52	0	1.02	2.26
100	4.43	4.56	3.50	0	1.57	3.14	0	1.01	2.15
150	2.69	3.39	3.19	0	1.32	2.69	-	-	-
200	2.57	3.25	3.10	0.02	1.12	2.38	-	-	-
250	1.58	2.84	3.19	0.02	1.19	2.72	-	-	-
300	0.50	2.16	2.84	0.01	1.32	2.45	-	-	-
500	0.27	1.23	1.58	0.01	0.61	1.32	-	-	-

power constraints, to disrupt rogue IoT networks as a defence mechanism. The results obtained from our framework may be used as a design and validation stage for a real implementation of the jammer, by allowing different scenarios and device settings to be examined.

5 | DISCUSSION AND FUTURE DIRECTIONS

Our case study results in Section 4 demonstrated the ability to perform testing of complex jamming scenarios with hardware-in-the-loop. We analysed three specific examples; however, the experimentation framework is capable of supporting numerous different site-specific jamming experiments. Various numbers and types of devices under test, jammers, physical environments, and levels of mobility may be considered. In addition to jamming, we have used the ray-tracing emulation system to examine Ultra-high-frequency (UHF) Radio Frequency Identification (RFID) for wireless IoT devices, thus highlighting the versatility of the system [65]. The overall workflow we described is suitable for other channel emulator systems and may be extended for compatible scenario development.

Ongoing efforts are focussed on (i) enabling remote access of the testbed for the research community, and (ii) integrating a PHY layer mitigation technique into DragonRadio. We have successfully used our Grid SDR Network Testbed for remote wireless research and education while additional development is required for wired studies. In particular, we are focussing on the interactive selection of emulation scenarios and reservation systems necessary to open to the community. Data sets may also be generated using the testbed and shared with the research community for use. The data files related to the case studies presented in Section 4 are provided in [66]. In addition, we are extending the real-time online RALA state selection implementation of the work in [67] to include jammer classification. Features discussed in [28] will be used with machine learning techniques to classify jammer types and fine-tune the learning parameters of the state selection. The experimentation

framework will be used to provide an array of jamming threats and wireless environments for the analysis.

6 | CONCLUSION

In this work, we have presented a framework for conducting wireless communication experiments under active jamming scenarios. Cyber-physical systems must be secure and reliable to support transportation, energy, healthcare, and other areas of society. The evaluation of wireless communication systems is often limited due to licencing and permit requirements, as well as recent closures due to a global pandemic. We used site-specific ray-tracing models with a large-scale channel emulator to evaluate physical hardware under complex jamming scenarios. System architecture and capabilities were discussed for devices under test, real-world jamming adversaries, channel modelling, and emulation. We presented a method of developing new and diverse test scenarios using the system. In particular, we examined case studies of (i) mitigating a constant jammer during UAV communication through the use of reconfigurable antennas, (ii) time scheduling around a periodic jammer for vehicular technologies, and (iii) the effect of different reactive jamming parameters as a defence mechanism against a rogue IoT network. Our results demonstrated the value of our experimentation framework through the evaluation of complex jamming scenarios with hardware-in-the-loop. We discussed the applications of the framework for additional studies and the ability to adapt the described workflow for other channel emulator systems for scenario development. Continuing efforts are focussed on enabling remote access of the testbed for the research community and the real-time SDR implementation of a jamming mitigation technique leveraging the PHY layer.

ACKNOWLEDGEMENTS

This material is based upon work supported by the National Science Foundation under grants CNS-1730140 and CCF-1717088, and the Department of Education Graduate

Assistance in Areas of National Need (GAANN) programme under award P200A180082.

CONFLICT OF INTEREST

None.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.4749668>, reference number 10.5281/zenodo.4749668.

ORCID

Marko Jacovic  <https://orcid.org/0000-0001-7823-0357>

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How to cite this article: Jacovic, M., et al.: Experimentation framework for wireless communication systems under jamming scenarios. *IET Cyber-Phys. Syst., Theory Appl.* 7(2), 93–111 (2022). <https://doi.org/10.1049/cps2.12027>